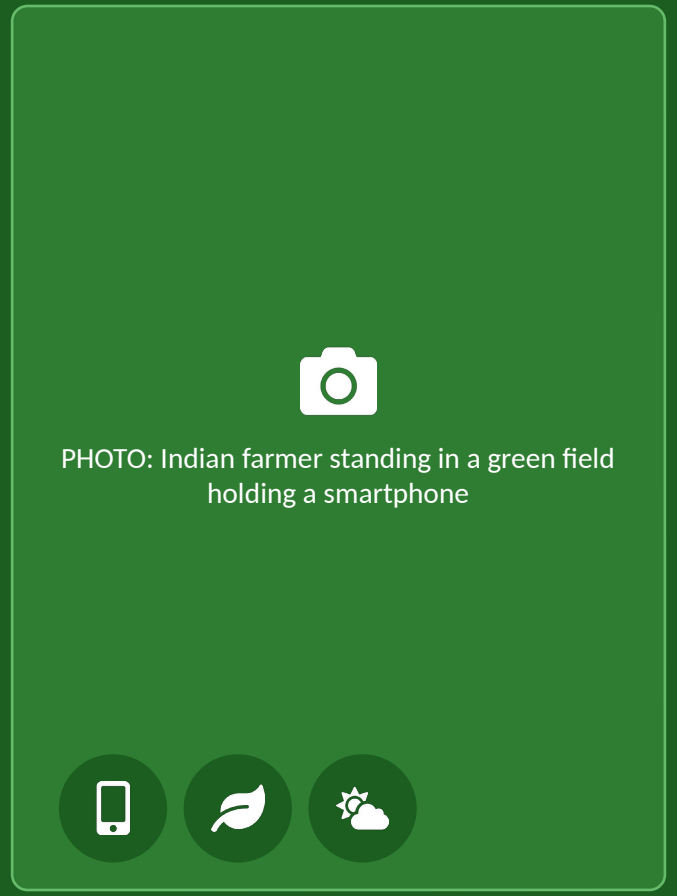
SMART CROP ADVISORY SYSTEM

For Small and Marginal Farmers

An Integrated Digital Platform for Crop-to-Market Support

Student Name: _____
Department: _____
College: _____
Project Guide: _____
Academic Year: _____



Introduction

- Crop selection and seed selection
- Weather awareness
- Irrigation planning
- Harvesting
- Marketing and selling
- Customer connectivity
- Transportation



The proposed digital platform brings multiple farmer-support services together in one place.

Problem Statement

PHOTO: Background photograph of Indian agricultural fields or farmers

Difficulty selecting suitable crops and seeds

Lack of timely weather information

Difficulty planning irrigation

Limited harvesting guidance

Lack of market-price and selling information

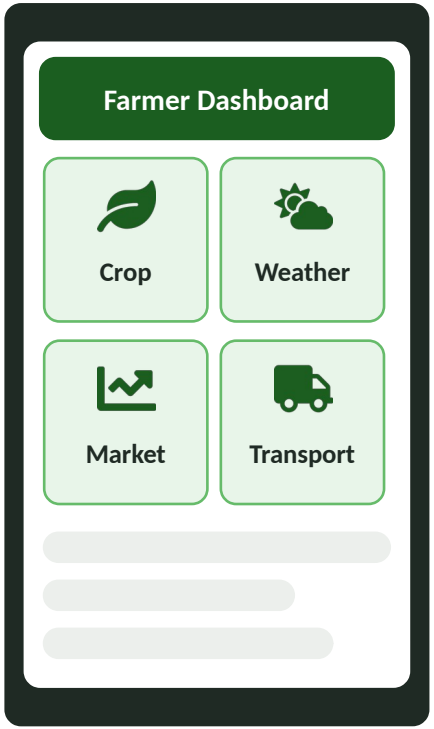
Difficulty connecting directly with customers

Transportation challenges

Agricultural information spread across different sources

Proposed System

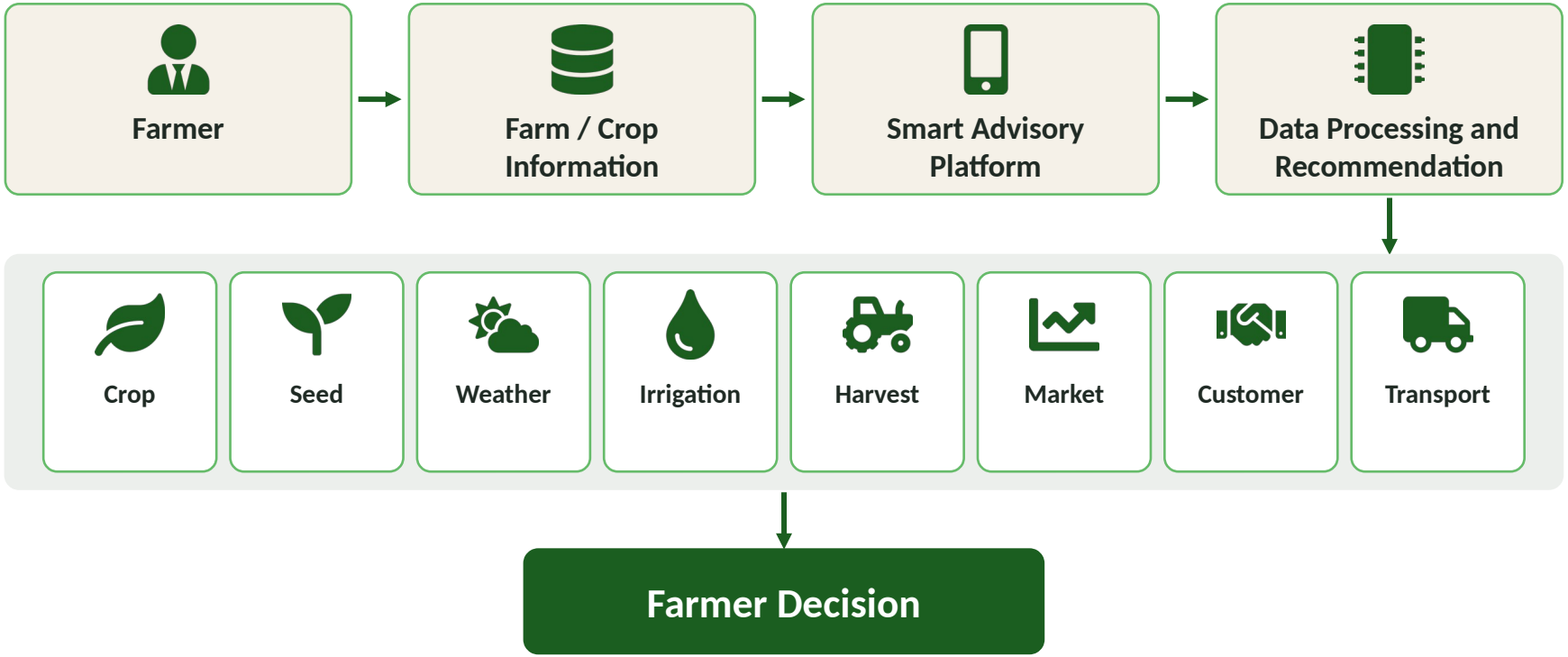
- Crop selection guidance
- Seed selection support
- Weather information
- Irrigation guidance



- Harvesting guidance
- Market information
- Customer connection
- Transportation support

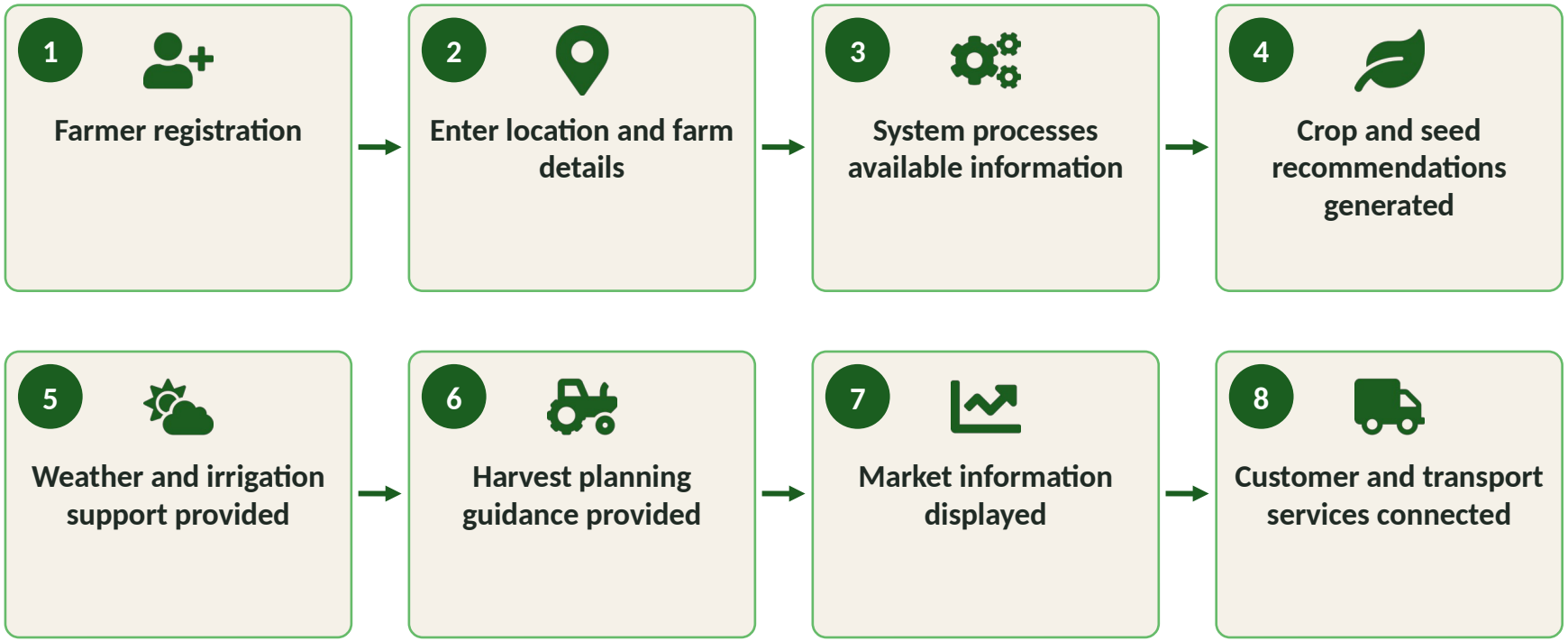
Conceptual mockup - not an implemented application

System Architecture



Proposed conceptual architecture - not an implemented system

Working Methodology

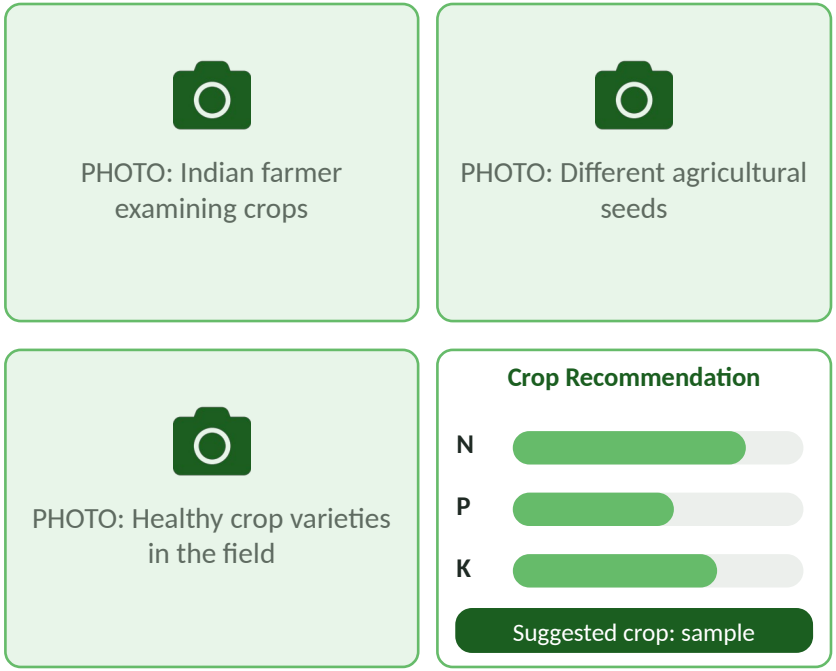


Steps 1-4 (top row) continue into steps 5-8 (bottom row)

Crop and Seed Recommendation

Nitrogen (N)	**Phosphorus (P)**
Potassium (K)	**Soil pH**
Temperature	**Humidity**
Rainfall	**Environmental parameters**

- Soil and climate data guide crop choice
- Helps match suitable seeds to local conditions
- Recommendations are proposed, not validated



Conceptual dashboard - illustrative only

Weather Information and Irrigation Support

Weather and Environmental Information → **Irrigation Planning** → **Farmer Decision**

- Weather awareness
- Rainfall information
- Temperature and humidity
- Irrigation planning
- Water management

PHOTO: Indian farm under changing weather, dark clouds and sun

PHOTO: Irrigation water flowing through crop rows

Weather (sample)

-- C -- % -- mm

Illustrative values

Soil moisture

Illustrative water management

Harvesting Guidance

Crop Monitoring → **Harvest Planning** → **Harvesting** → **Post-Harvest Handling**

- Harvest planning
- Identifying appropriate harvesting activities
- Harvested crop handling
- Preparation for selling and transportation

PHOTO: Indian farmers harvesting mature paddy

PHOTO: Harvested grains, vegetables or crop samples

Guidance only - automatic maturity prediction is a future enhancement

Market Information

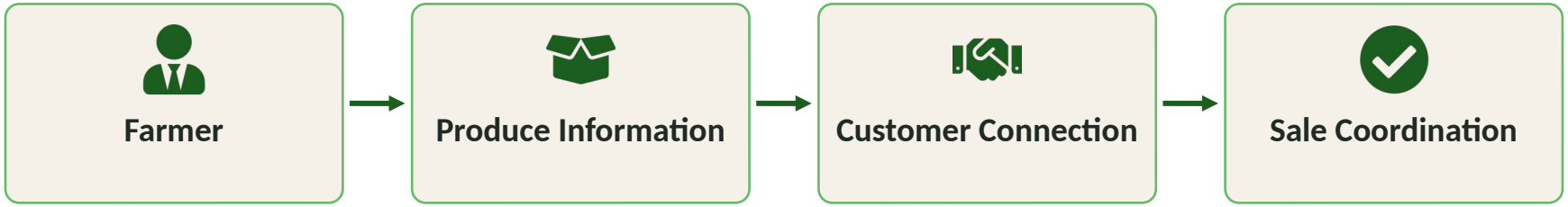
- Access to market information
- Awareness of selling options
- Support for market-related decisions
- Preparation for selling harvested produce



Crop category	Price (Rs/quintal)
Cereals	--- *(illustrative)*
Vegetables	--- *(illustrative)*
Pulses	--- *(illustrative)*
Fruits	--- *(illustrative)*

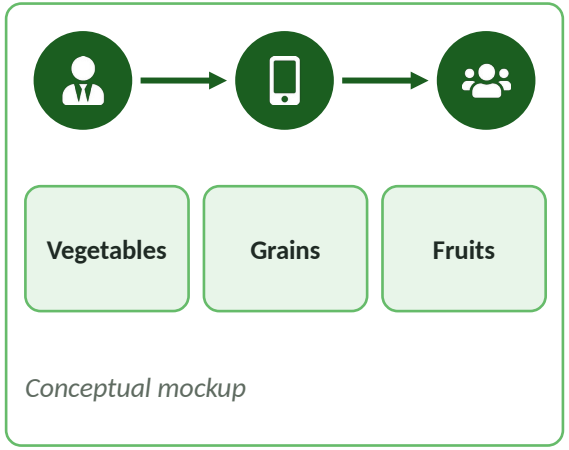
Conceptual dashboard - placeholder fields, no real prices

Customer Connection



- Connect farmers and potential customers
- Display available produce
- Support direct communication
- Help coordinate produce sales

PHOTO: Indian farmer handing fresh vegetables to a customer



Conceptual mockup

Transportation Support

Farm → **Collection Point** → **Market / Customer**

- Transportation coordination
- Produce movement from farms
- Delivery planning
- Links between collection points, markets and customers

PHOTO: Small transport truck carrying fresh produce on a rural Indian road

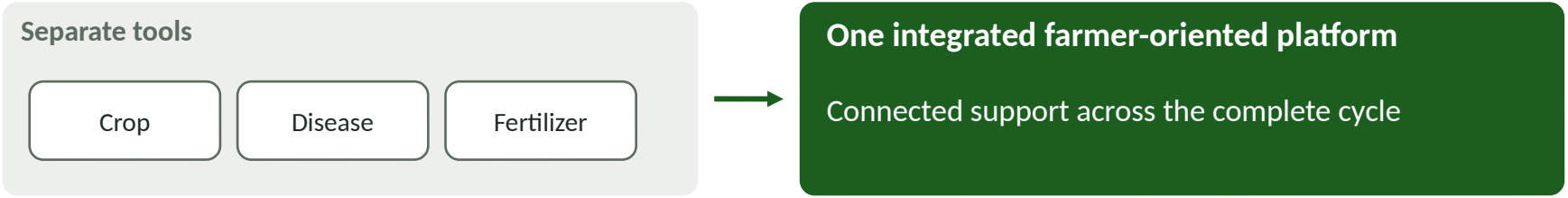
PHOTO: Crates of harvested vegetables

Conceptual module - no live tracking or route optimisation claimed

Existing Research and Research Gap

- Crop recommendation (NPK values)
- Soil pH
- Temperature and humidity
- Rainfall and environmental parameters
- Plant disease detection
- Fertilizer recommendation
- Yield prediction
- Multilingual advisory platforms

Research gap: many systems cover only one or a few functions.



Crop Selection → Cultivation → Irrigation → Harvesting → Marketing → Selling → Transportation

Feasibility and Scalability

Technical Feasibility

Could use web or mobile technologies, databases, APIs and machine learning models.

Economic Feasibility

A software-based platform may reduce the need for initial physical infrastructure.

Operational Feasibility

A simple interface could improve accessibility for farmers.

Future Scalability

Potential additions: AI, real-time weather and market data, regional languages, voice interaction, government services and IoT monitoring.

Feasibility is a proposal; no implementation or cost analysis is available.

Expected Benefits and Future Scope

Expected Benefits

- Better-informed crop selection
- Timely agricultural information
- Improved irrigation planning
- Better harvest planning
- Easier access to market information
- Direct farmer-customer connection
- Easier transportation coordination
- Multiple services through one platform

Future Scope (not yet implemented)

- AI-powered personalized crop recommendations
- Machine learning for crop and market prediction
- IoT-based soil and environmental monitoring
- Computer vision for plant disease identification
- Regional languages and voice interaction
- Real-time market-price information
- Government agricultural scheme information
- Smart irrigation control

Smart farming concept



Conclusion and References

- The Smart Crop Advisory System is a proposed integrated digital platform supporting small and marginal farmers throughout the agricultural cycle.
- Core modules: crop selection, seed selection, weather information, irrigation, harvesting, market information, customer connectivity and transportation.
- It provides a foundation for future AI-, IoT- and data-driven agricultural services.

References

1. Smart Crop Advisory System for Small and Marginal Farmers, 2026.
2. Smart Crop Advisory System: A Machine Learning Approach to Precision Crop Recommendation Using Soil and Climatic Parameters, 2026.
3. An IoT-enabled AI system for real-time crop prediction using soil and weather data in precision agriculture, Smart Agricultural Technology, 2025.
4. A Review on AI-Driven Smart Crop Advisory Systems for Small and Marginal Farmers, 2026.
5. KRISHI.AI: an explainable machine learning framework for data-driven crop recommendation, Scientific Reports, 2026.

THANK YOU

Smart Crop Advisory System for Small and Marginal Farmers



PHOTO: Indian agricultural landscape with green crops and a farmer